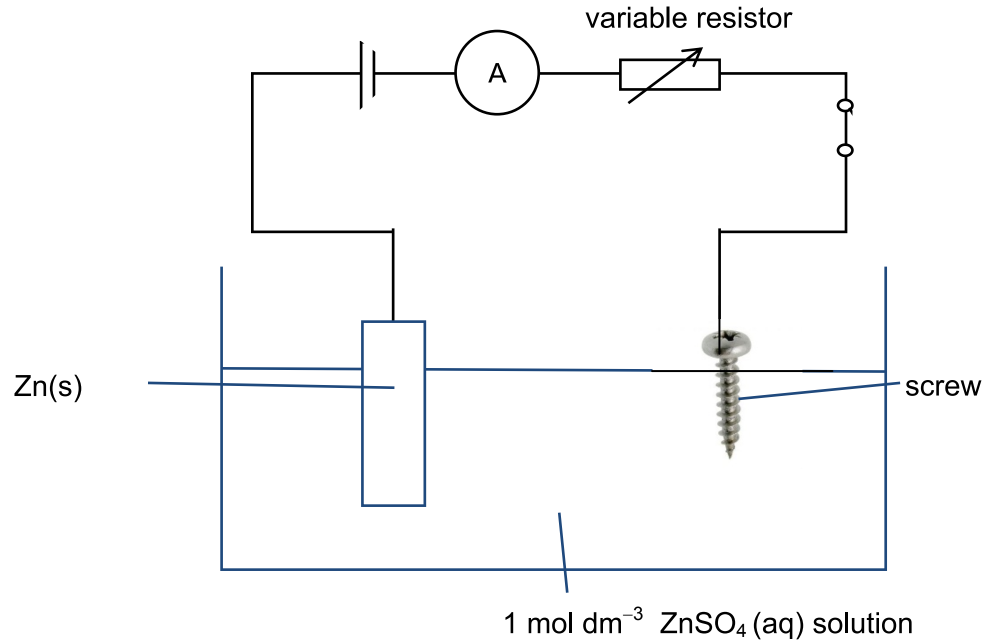
1 (a) Cathode: $Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$
Anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

(b)



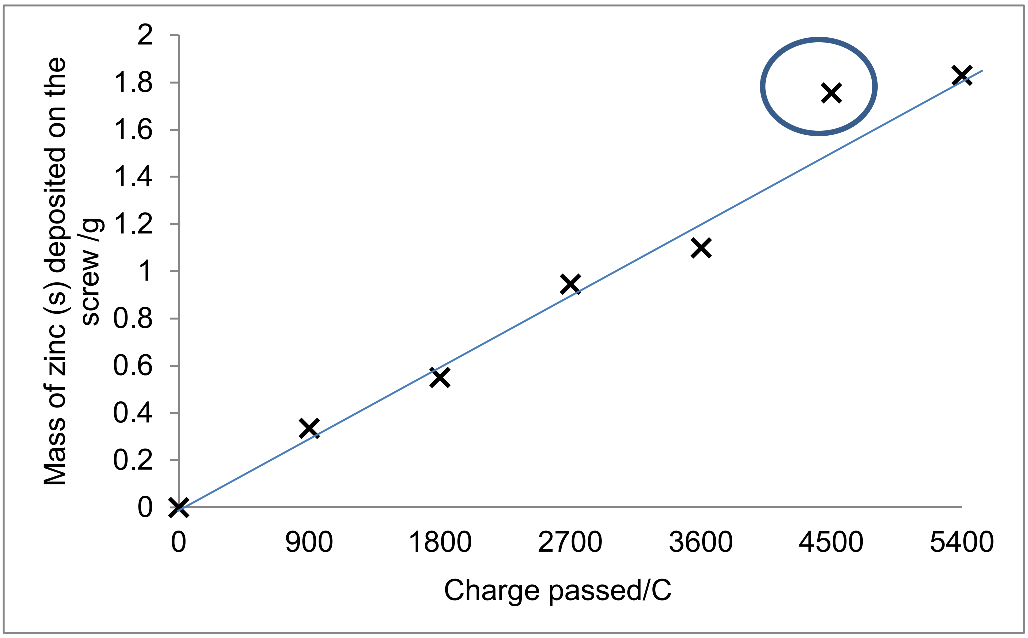
(c)

Amount of Zn^{2+} = $\dfrac{500}{1000} \times 1$ = 0.500 mol

Mass of $ZnSO_4$ solid required = 0.500 x 161.5 = 80.75 g

- Using a weighing balance, weigh out 80.75g of zinc sulfate solid into a 100cm^3 weighing beaker.
- Pour distilled water into a 250cm^3 (OR 500cm^3) beaker, and add in the solid to dissolve in small amount. (Do not accept pouring distilled water into the large amount of solid as the dissolving process would take a long time.)
- Stir the solution with a glass rod (or stirrer) until all the solid has dissolved.
- Rinse the weighing beaker with distilled water and allow the washings to drip into the beaker. Rinse for a few times.
- Pour the concentrated solution using the aid of a glass rod into a clean 500 cm^3 volumetric flask using a funnel.
- Rinse the beaker a few times and add the washings into the volumetric flask.
- Carefully make up the mark with distilled water using a dropper.
- Stopper the standard flask and shake well to obtain a homogenous solution.

(d)



(e) Correct anomalous point circled.
The anomaly was higher than usual because the screw was not totally dried/ the water was not fully removed when the mass was measured.

(f) $n_e \times F = Q$

$n_e = \frac{Q}{F}$

$Zn^{2+} + 2e \rightarrow Zn$

$2 \times \frac{massofZn}{Mr} = \frac{Q}{F}$

Mass of Zn = $(\frac{M_r}{2F}) \times Q = (\frac{65.4}{2F}) \times Q$

$= (\frac{32.7}{F}) \times Q$

Gradient = $(\frac{32.7}{F})$ [

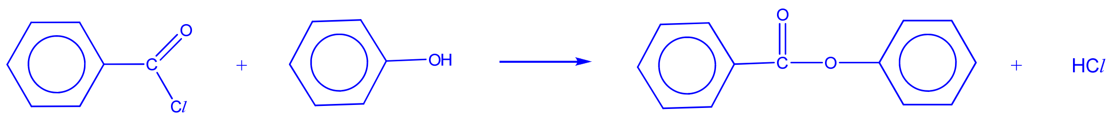
$3.38 \times 10^{-4} = (\frac{32.7}{F})$

F= 96700 C mol^{-1} (3 s.f.)

(g) It is difficult to keep the current constant
OR
Not all the zinc is deposited on the cathode and some are dispersed in the electrolyte as fine powder.

2 (a) NaOH will react with the phenol to produce phenoxide which is a better /stronger nucleophile than phenol.

(b)



(c) Mr of phenol = 6(12.0) + 5.0 + 16.0 + 1.0 = 94.0

Amount of phenol = $\frac{2.0}{94.0} = 0.0213\text{mol}$

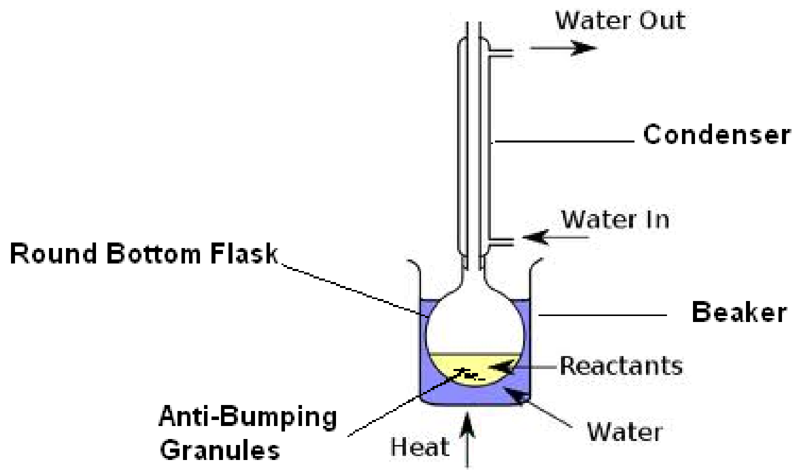
Mr of benzoyl chloride = 6(12.0) +5.0+12.0+16.0+35.5 = 140.5

Concentration of benzoyl chloride in mol dm^{-1} = $\frac{58}{140.5} = 0.4128\text{mol dm}^{-3}$

Amount of benzoyl chloride = $\frac{0.4128}{1000} \times 40 = 0.0165\text{mol}$

Phenol is in excess.

(d)



(e) For reactions involving organic molecules to occur, their covalent bonds need to be broken. Heating the reaction mixture for a sufficient length of time is necessary to provide the energy required for the endothermic bond breaking to take place.

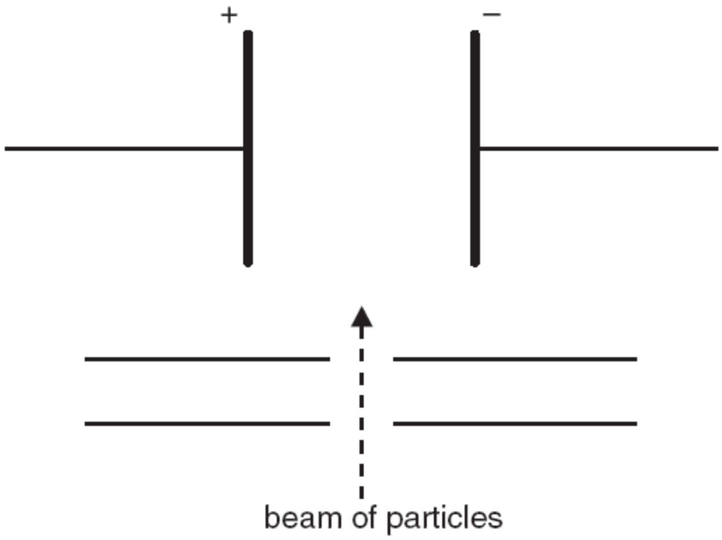
(f) To remove impurities and purify the products.

(g) Phenol do not react with benzoic acid as it is a weaker nucleophile as compared to alcohol.
The reaction using benzoyl chloride goes to completion.
The reaction using benzoyl chloride is rapid.

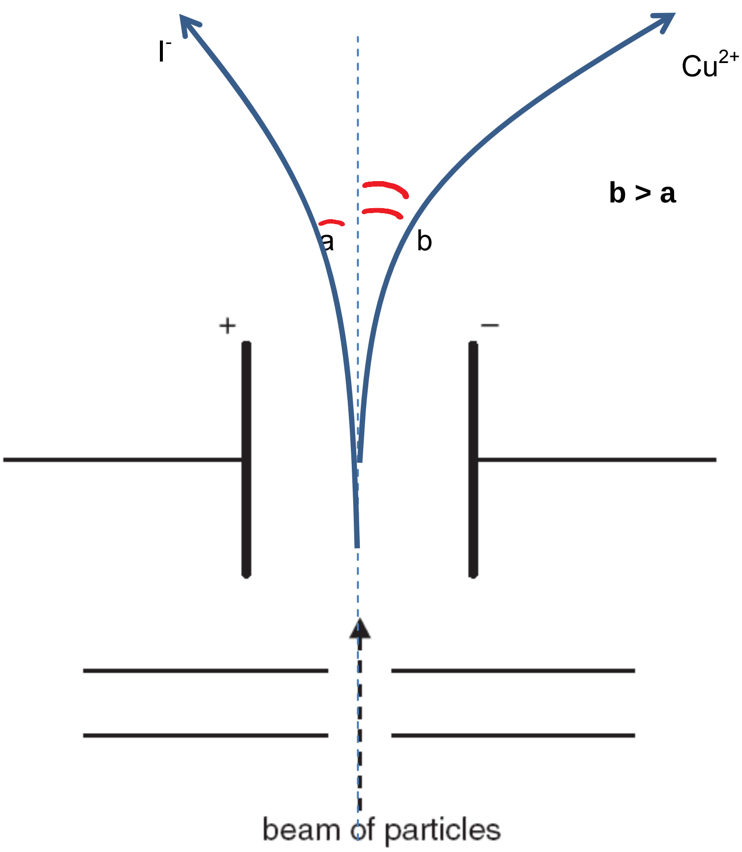
3 (a) Percentage abundance of the isotopes in this sample differs from what is normally

obtained OR There are more than 2 types of isotopes of copper.

(b)



(i)



(ii)

deflection angle = $k\dfrac{\text{charge}}{\text{mass}}$ → 7 = $k\left(\dfrac{2}{63}\right)$ → k = 220.5

The value of k for I^- in the same electric field is 220.5.

Hence, deflection angle of I^- = $220.5\left(\dfrac{1}{127}\right)$ = 1.736 = -1.7 ^{0}(1 dp)

[4]

(c) B is P and C is Si

Chlorides (PCl_3 or PCl_5) and oxides of P (P_4O_6 or P_4O_{10}) hydrolyses in water to give acidic solutions.

$$P_4O_6(s) + 6H_2O(l) \longrightarrow 4H_3PO_3\,(aq)$$

OR

$$P_4O_{10}(s) + 6H_2O(l) \longrightarrow 4H_3PO_4\,(aq)$$

$$PCl_3(l) + 3H_2O(l) \longrightarrow H_3PO_3(aq) + 3HCl(aq)$$

OR

$$PCl_5(s) + 4H_2O(l) \longrightarrow H_3PO_4(aq) + 5HCl(aq)$$

Oxide of C (SiO_2) has a giant covalent structure and thus have high melting point . It is an acidic oxide as it reacts with concentrated NaOH

$$SiO_2(s) + 2\,OH^-(conc) \longrightarrow SiO_3^{2-}(aq) + H_2O(l)$$

(d) (i) A solution of Cr^{3+} is acidic in nature due to the high charge density of the cation and is able to polarise the electron cloud of water molecule.

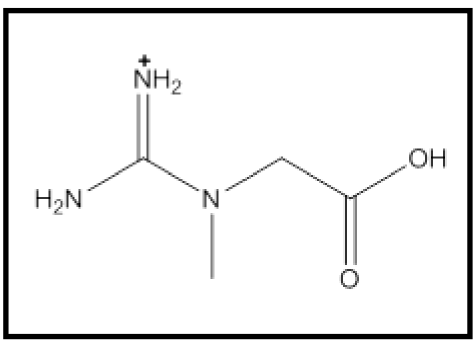
$$[Cr(H_2O)_6]^{3+}(aq) + H_2O(l) \rightleftharpoons [Cr(OH)(H_2O)_5]^{2+}(aq) + H_3O^+(aq)$$

(d) (ii) chromium(III) hydroxide OR $Cr(OH)_3$ OR $[Cr(OH)_3(H_2O)_3]$

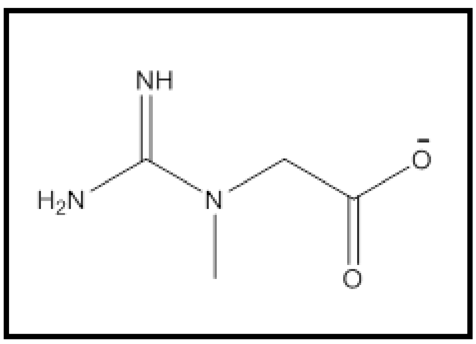
(iii) $[Cr(OH)_6]^{3-}$ OR $[Cr(OH)_4]^-$

(iv) Ligand exchange reaction has occurred. Different ligands cause different splitting of the d-orbitals and hence different energy gaps, which correspond to different wavelengths of visible light being absorbed. Hence different colours are observed.

4 (a) (i)

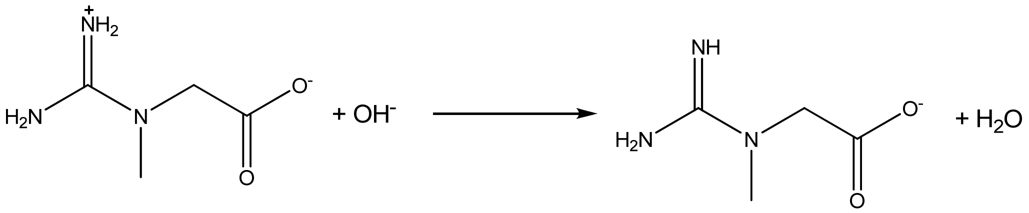


pH 6

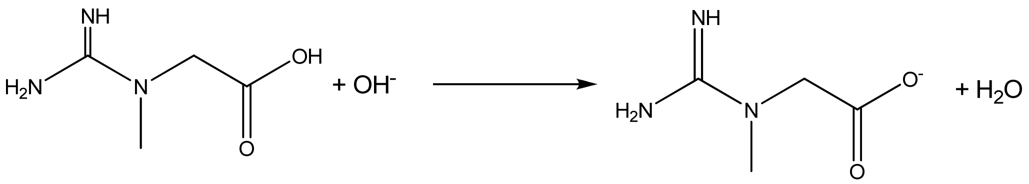


pH 12

(ii)



OR



(iii) No. Creatine is not optically active as it does not have any chiral carbon.

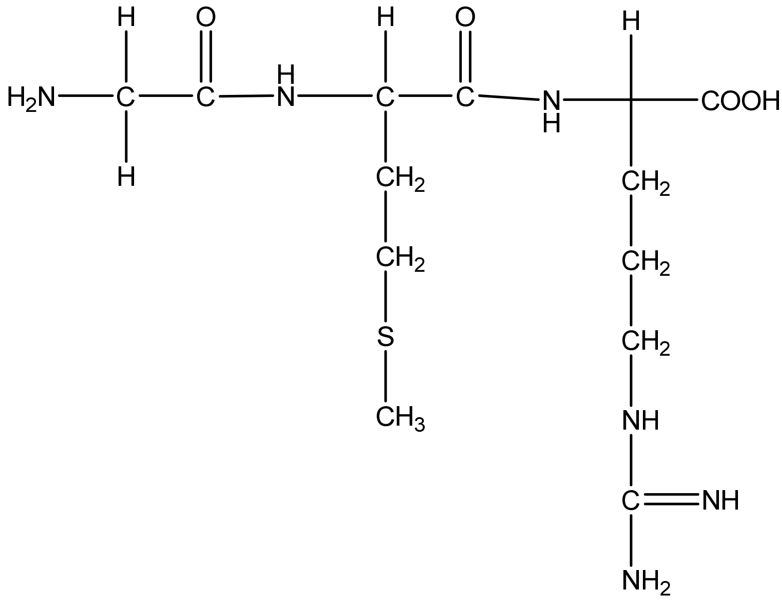
(iv) Test: Add neutral $FeCl_3$ to both compounds separately at room temperature.
Observation for creatine: No violet colouration.
Observation for tyrosine: Violet colouration formed.

Or
Test: Add aqueous bromine at room temperature to both compounds separately at room temperature.
Observation for creatine: Orange bromine remains orange.
Observation for tyrosine: Orange bromine decolourises, white precipitate and steamy white fumes produced.

Or
Test: Add liquid bromine at room temperature to both compounds separately at room temperature.
Observation for creatine: Reddish brown bromine remains reddish brown.
Observation for tyrosine: Reddish brown bromine decolourises, white precipitate and steamy white fumes produced.

Or
Test: Add $KMnO_4$ with H_2SO_4 to both compounds separately and heat.
Observation creatine: The solution remains purple.
Observation for tyrosine: The purple solution decolourises and white precipitate formed.

(b) (i)

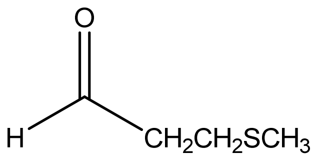


(ii) Condensation

(iii) Denaturation is the disruption of the shape of protein without changing its primary structure but results in loss of biological activity.

(iv) The weak R-group interaction such as the van der Waals' interaction and hydrogen bonds is disrupted by the heat.

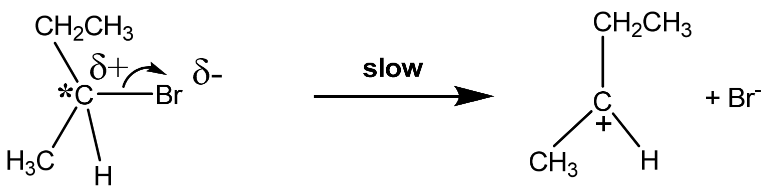
(c) (i)

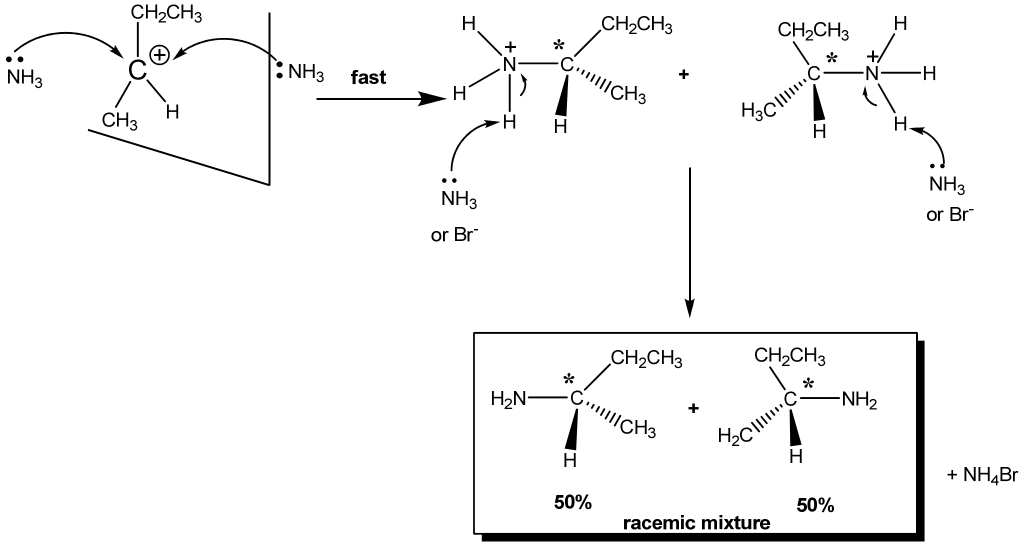


(ii) Step I : Nucleophilic addition
Step II : Elimination

(iii) Reagent: aqueous HC*l* / aqueous H_2SO_4
Condition: heat with reflux

(iv) Nucleophilic substitution S_N1 (If student draw S_N2, NO marks)





5 (a) (i) pH = - log ($\sqrt{K_a[\text{acid}]}$)

pH = - log ($\sqrt{1.41 \times 10^{-3}[0.100]}$)

= 1.93

(ii) $HO_2CCH_2CO_2H + Ba(OH)_2 \rightarrow Ba(O_2CCH_2CO_2) + 2H_2O$

Amount of malonic acid = $\frac{0.100}{1000} \times 25.0 = 0.00250$ mol

Since amount of malonic acid : amount of $Ba(OH)_2$ = 1: 1

Amount of $Ba(OH)_2$ = 0.00250 mol

Volume of $Ba(OH)_2$ needed = $\frac{1000}{0.0625} \times 0.00250 = 40.0\,cm^3$

(iii) $HO_2CCH_2CO_2H$ and $HO_2CCH_2CO_2^-$ in equal concentration = first maximum buffer capacity

pH = pK_{a1}

= - log (1.41 x 10^{-3})

= 2.85

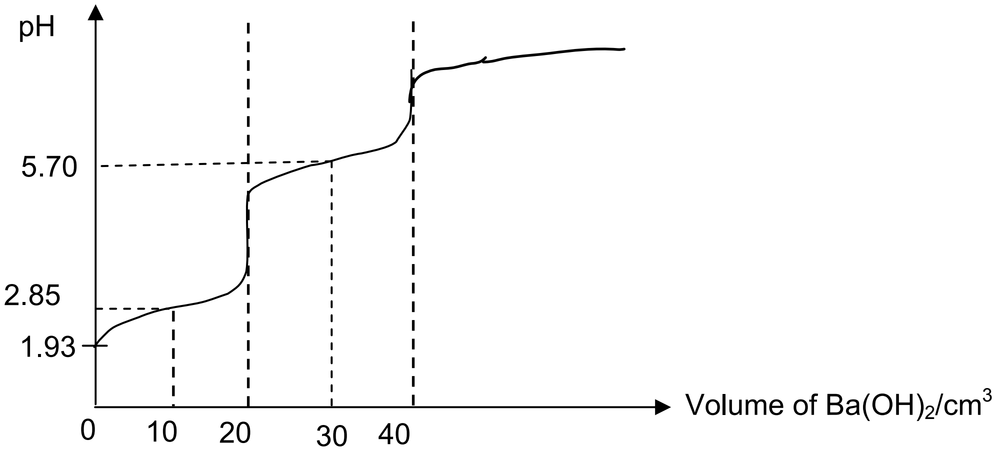
(iv) $HO_2CCH_2CO_2^-$ and $^-O_2CCH_2CO_2^-$ in equal concentration = second maximum buffer capacity

pH = pK_{a2}

= -log (2.00 x 10^{-6})

= 5.70

(v)



(b) (i) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$

Amount of HCl = $\frac{20}{1000} \times 0.2 = 0.004000\text{mol}$

Amount of Na_2CO_3 = 0.004000/2 = 0.002000 mol

Therefore,

Concentration of Na_2CO_3 in mol dm^{-3} = $0.002 \times \frac{1000}{25} = 0.0800\text{moldm}^{-3}$

(ii) Number of moles of Na_2CO_3 in 250cm^3 = 0.002000 x 10 = 0.0200 mol

Mr of $Na_2CO_3.xH_2O$ = 2 x 23.0 + 12.0 + 3 x 16.0 + x (18.0)

= 106 + 18x

$$\frac{5.00}{106 + 18x} = 0.0200$$

2.12 + 0.36x = 5

0.36x = 2.88

x = 8